The Gravitational Refractive Index as Petz Recovery Fidelity: Temporal Asymmetry from Spacetime Geometry

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In the companion paper (Paper I), we established that the temporal asymmetry parameter $\tau = 1 - F$ quantifies the arrow of time through the Petz recovery fidelity bound $F \geq e^{-\Sigma/2}$. Here we connect this framework to gravity. In weak gravitational fields, the coordinate speed of light satisfies $c_{\rm eff}/c = 1 + \Phi/c^2 + O(\Phi^2/c^4)$, which exponentiates to $c_{\rm eff}/c = e^{\Phi/c^2}$—exact through post-Newtonian order. We identify this with the Petz bound: $c_{\rm eff}/c = e^{-\Sigma_{\rm grav}/2} = F_{\rm bound}$, where $\Sigma_{\rm grav} = 2|\Phi|/c^2$. Light slowing in a gravitational field *is* information loss; the gravitational refractive index is the inverse of the recovery fidelity. The exponential metric $g_{00} = -e^{-r_{\rm s}/r}$ corresponds to the *lower bound geometry*—the metric where recovery fidelity equals the JRSWW bound $F = e^{-\Sigma_{\rm grav}/2}$. At a Schwarzschild event horizon, $\tau \to 1$ (complete information loss), while in the exponential metric no horizon forms. We show that Pikovski's gravitational decoherence mechanism is naturally a Petz recovery problem, and that several results in black hole physics—the Page curve, Hayden–Preskill protocol, entanglement wedge reconstruction—are instances of the τ bound. Three complementary physical frameworks (algebraic QFT, quantum optics, gravitational thermodynamics) yield the same identification, confirming its structural robustness.

INTRODUCTION

In the companion paper (Paper I [2]), we established three results. First, the Petz recovery map $\mathcal{R}$ is the unique retrodiction functor satisfying Bayesian consistency [3]. Second, its fidelity obeys the universal bound

$$F(\rho,\ \mathcal{R}\circ\mathcal{N}(\rho)) \ \geq\ e^{-\Sigma/2}\,, \tag{1}$$

where Σ is the entropy production of the quantum channel $\mathcal{N}$ [4, 5, 34]. This bound has recently been verified experimentally in ion-trap and NMR platforms [35, 36].[1] Third, the temporal asymmetry parameter $\tau \equiv 1 - F \in [0, 1]$ quantifies the arrow of time: $\tau = 0$ for unitary evolution (perfect retrodiction); $\tau > 0$ for open processes that leak information to an environment.

The equivalence chain established in Paper I reads:

$$\tau = 0 \Leftrightarrow \Sigma = 0 \Leftrightarrow \text{perfect recovery} \Leftrightarrow \text{no time arrow}. \tag{2}$$

What does τ have to do with gravity?

In general relativity, the coordinate speed of light varies with gravitational potential. This is not a modification of GR but a standard consequence of the metric structure, recognized since Einstein's 1911 analysis [7] and placed on rigorous footing by Dicke's refractive-medium formulation [8]. In weak fields, the linearized coordinate speed is $c_{\rm eff} \approx c(1 + \Phi/c^2)$ [KNOWN]. The natural exponentiation,

$$c_{\rm eff} = c\, e^{\Phi/c^2}\,, \tag{3}$$

where $\Phi < 0$ is the Newtonian potential, agrees with the Schwarzschild metric through post-Newtonian order and defines the exponential (Papapetrou–Yilmaz) metric at all orders [9–11]. The two metrics diverge at $O(r_{\rm s}/r)^3$; we discuss this in Sec. .

The central observation of this paper is that Eq. (3) is *identical in form* to the Petz recovery fidelity bound (1) [NEW SYNTHESIS]. Defining the gravitational entropy production $\Sigma_{\rm grav} \equiv 2|\Phi|/c^2 = r_{\rm s}/r$ (where $r_{\rm s} = 2GM/c^2$), we obtain:

$$\boxed{\frac{c_{\rm eff}}{c} \ =\ e^{-\Sigma_{\rm grav}/2} \ =\ F_{\rm bound}\,.} \tag{4}$$

The effective speed of light *is* the Petz recovery fidelity bound—where the JRSWW bound defines the geometry. This identification means the exponential metric is the *lower bound geometry*: by Theorem 1 in the Supplemental Material, exact saturation is impossible for $\Sigma_{\rm grav} > 0$, so the actual recovery fidelity strictly exceeds $e^{-\Sigma_{\rm grav}/2}$. The exponential metric is the geometry where F equals this lower bound, and any deviation from it is an empirical question [IDENTIFICATION, not hypothesis]. The gravitational refractive index $n_{\rm grav} = c/c_{\rm eff} = e^{+\Sigma_{\rm grav}/2} = 1/F_{\rm bound}$ is the *inverse* of the recovery fidelity.

The Petz map has already emerged as a central tool in black hole physics. Chen, Penington, and Salton [25] showed that entanglement wedge reconstruction in AdS/CFT is implemented by the Petz map. Cotler *et al.* [26] demonstrated that the Hayden–Preskill protocol is an instance of universal recovery. Our identification (4) extends these connections: not only does the Petz map reconstruct bulk operators, but the *metric itself* encodes the recovery structure.

The paper is organized as follows. Section establishes the exponential identification and provides numerical verification. Section reinterprets Pikovski's gravitational decoherence as a Petz recovery problem. Section shows that multiple results in black hole physics are instances of the τ bound. Section delineates what τ can and cannot

explain. Section discusses open questions and outlook.

THE EXPONENTIAL IDENTIFICATION

From GR to recovery fidelity

The starting point is two established results.

From GR [KNOWN].—In a static gravitational field with Newtonian potential $\Phi = -GM/r$, the g_{00} component of the metric determines the coordinate speed of light: $c_{\rm eff}/c = \sqrt{-g_{00}}$ [KNOWN]. In weak fields, $g_{00} = -(1 + 2\Phi/c^2) + O(\Phi^2/c^4)$, so $c_{\rm eff}/c = 1 + \Phi/c^2 + O(\Phi^2/c^4)$ [KNOWN]. The exponentiation $c_{\rm eff}/c = e^{\Phi/c^2}$ [7, 8] agrees with GR through post-Newtonian order and defines the exponential (Papapetrou–Yilmaz) metric at all orders [9, 10]. At $O(r_{\rm s}/r)^3$ it diverges from Schwarzschild—see Sec. .

From quantum information [KNOWN].—The Petz recovery fidelity satisfies $F \geq e^{-\Sigma/2}$, where Σ is the entropy production (QRE decrease) [4, 5].

The identification [NEW].—We define the gravitational entropy production:

$$\Sigma_{\rm grav} \equiv \frac{2|\Phi|}{c^2} = \frac{r_{\rm s}}{r}\,, \tag{5}$$

which is the unique dimensionless gravitational compactness parameter. Then Eq. (3) becomes $c_{\rm eff}/c = e^{-\Sigma_{\rm grav}/2} = F_{\rm bound}$: the coordinate speed of light equals the Petz recovery fidelity bound.

The physical content is: *light slowing in a gravitational field is information loss.* The "cost of retrodiction" for a signal climbing through potential difference $|\Phi|$ is quantified by $\Sigma_{\rm grav}$, and the maximum recoverable fidelity is $e^{-\Sigma_{\rm grav}/2}$.

We emphasize from the outset what this identification does *not* provide: it does not predict dark matter abundances, dark energy values, or allow reconstruction of the metric from τ alone (see Sec. for a complete accounting).

Three complementary interpretations

Three complementary physical frameworks yield the same identification $\Sigma_{\rm grav} = -\ln(-g_{00})$, confirming its structural robustness [KNOWN individually; NEW as convergent package]:

(a) *Modular flow (algebraic QFT).*—In the entanglement first-law regime on a Schwarzschild background, the quantum relative entropy at radius r exceeds that at infinity due to the Tolman blueshift. The fractional QRE loss is $\Delta D/D = r_{\rm s}/r$ [17].

(b) *Quantum channel model (quantum optics).*—Modeling gravitational redshift as a thermal attenuator bosonic channel with intensity transmissivity $\eta = -g_{00}$ [18], the channel's entropy production is $\Sigma = -\ln\eta = -\ln(-g_{00})$. Explicit Kraus operators can be constructed via the Stinespring dilation of this channel [37].

(c) *Gravitational Landauer principle (thermodynamics).*—The Tolman-modified cost of erasing one bit at radius r exceeds the cost at infinity by $1/\sqrt{-g_{00}}$ [19, 21]. The dimensionless excess erasure cost gives $\Sigma_{\rm grav} = -\ln(-g_{00})$.

We stress that these are three *interpretations* of the same mathematical structure, not three independent derivations. All share the common input g_{00} from GR and output $\Sigma_{\rm grav} = -\ln(-g_{00})$. Their convergence across distinct physical frameworks (algebraic QFT, quantum optics, thermodynamics) is evidence of robustness, not of independent confirmation.

TABLE I. Gravitational entropy production, Petz fidelity bound, and temporal asymmetry for representative environments. $\Sigma_{\rm grav} = r_{\rm s}/r$; $F_{\rm bound} = e^{-\Sigma_{\rm grav}/2}$; $\tau \leq 1 - F_{\rm bound}$. All values are exact in the weak-field regime ($\Sigma_{\rm grav} \ll 1$). The white dwarf assumes $M = 0.6\,M_\odot$, $R = 8000$ km; the neutron star assumes $M = 1.4\,M_\odot$, $R = 12.5$ km. *For Schwarzschild, $\Sigma_{\rm grav}^{\rm Schw} = -\ln(1 - r_{\rm s}/r) \to \infty$ at the horizon; the exponential metric has no horizon ($\Sigma_{\rm grav} = 1$ at $r = r_{\rm s}$, $F_{\rm bound} = 0.61$).

Environment	$\Sigma_{\rm grav}$	$F_{\rm bound}$	τ
Earth surface	1.4×10^{-9}	$1 - 7 \times 10^{-10}$	$\leq 7 \times 10^{-10}$
Solar surface	4.2×10^{-6}	$1 - 2.1 \times 10^{-6}$	$\leq 2.1 \times 10^{-6}$
White dwarf	2.2×10^{-4}	0.99989	$\leq 1.1 \times 10^{-4}$
Neutron star	0.33	0.847	≤ 0.153
Schw. horizon*	$\to \infty$	$\to 0$	$\to 1$

Numerical verification

Table I lists $\Sigma_{\rm grav}$, $F_{\rm bound}$, and τ for representative gravitational environments.

The gravitational refractive index $n_{\rm grav} = 1/F_{\rm bound} = e^{+\Sigma_{\rm grav}/2}$ has a transparent operational meaning: it is the factor by which retrodiction degrades when climbing out of a gravitational well. At the Earth's surface, $n_{\rm grav} = 1 + 7 \times 10^{-10}$: gravity introduces an imperceptible "refractive delay" in information recovery. At a neutron star surface, $n_{\rm grav} \approx 1.18$: an 18% degradation. At a Schwarzschild horizon ($\Sigma_{\rm grav}^{\rm Schw} \to \infty$), $n_{\rm grav} \to \infty$ and $\tau \to 1$: complete information loss. In the exponential metric, $\Sigma_{\rm grav} = 1$ at $r = r_{\rm s}$, giving $n_{\rm grav} = e^{1/2} \approx 1.65$ and $\tau \leq 0.39$—no complete loss occurs.

Weak-field exactness and strong-field caveat

The identification (4) is *exact* in weak fields ($\Sigma_{\rm grav} \ll 1$), where the exponential metric and the Schwarzschild metric agree through second post-Newtonian order: $g_{00}^{\rm exp} = g_{00}^{\rm Schw} + O((r_{\rm s}/r)^3)$. Both metrics yield PPN parameters $\gamma = \beta = 1$ exactly [11, 12].

In strong fields ($\Sigma_{\rm grav} \gtrsim 1$), the exponential and Schwarzschild metrics diverge. The exponential metric has no event horizon and has been classified as a traversable wormhole [13]. Whether nature selects the Schwarzschild metric (standard GR) or the exponential metric (bound geometry) is an empirical question, with distinguishing predictions at the $\sim 5\%$ level in black hole shadow size and quasinormal mode spectra [37]. This strong-field ambiguity does not affect the weak-field results of Secs. –, which depend only on $\Sigma_{\rm grav} \ll 1$.

Proposition 1 (Characterization of the exponential metric). *Among static, spherically symmetric metrics $g_{00} = -f(r)$, the exponential metric $f(r) = e^{-r_{\rm s}/r}$ is the unique solution for which $\Sigma_{\rm grav} \equiv -\ln(-g_{00}) = r_{\rm s}/r$ is exactly linear in $r_{\rm s}/r$. The Schwarzschild metric gives $\Sigma_{\rm grav}^{\rm Schw} = -\ln(1 - r_{\rm s}/r) = r_{\rm s}/r + \frac{1}{2}(r_{\rm s}/r)^2 + \cdots$, which is linear only to leading order.*

We do not derive this metric from an action principle; rather, we identify it as the metric that exactly linearizes the QRE–gravity correspondence (4).

GRAVITATIONAL DECOHERENCE AS PETZ RECOVERY

Pikovski *et al.* [14] showed that gravitational time dilation causes universal decoherence of composite quantum systems. A particle in a superposition of heights Δh in Earth's gravity acquires a relative phase between its internal energy eigenstates at rate $\Delta E\, g\, \Delta h/(\hbar c^2)$, leading to decoherence when the internal degrees of freedom are traced out.

We observe that this mechanism has the *formal structure* of a Petz recovery problem, though it is fundamentally distinct from the gravitational channel $\Sigma_{\rm grav}$.

The channel.—Define the gravitational decoherence channel as the partial trace over internal degrees of freedom:

$$\mathcal{N}_{\rm grav} = {\rm Tr}_{\rm int}\,. \tag{6}$$

This is structurally identical to the quantum eraser: tracing out "which-path" information (here, the internal clock) destroys coherence in the complementary basis (here, position). For a qubit internal clock, this produces a dephasing channel $\mathcal{N}_p(\rho) = \frac{1+p}{2}\,\rho + \frac{1-p}{2}\,\sigma_z \rho\, \sigma_z$ on the position degree of freedom, with coherence parameter $p = |\cos(\Delta E\, g\, \Delta h\, t\,/\, 2\hbar c^2)|$.

(Three key distinctions between the Pikovski channel and $\Sigma_{\rm grav}$ are detailed in Supplement S6: the Pikovski entropy is probe-dependent, has no finite supremum, and dephases in the position basis rather than the energy basis.)

The Petz recovery.—Reversing the decoherence requires reconstructing the internal-state correlations from the external state alone. The Petz map $\mathcal{R}_\sigma \circ \mathcal{N}_{\rm grav}$ is the optimal such reconstruction. Explicit calculation [37] shows that, for the maximally mixed reference $\sigma = I/2$, the Petz recovery *is the dephasing channel itself*: $\mathcal{R}_{I/2} = \mathcal{N}_p$. This is because dephasing is self-adjoint ($\mathcal{N}_p^\dagger = \mathcal{N}_p$) and $I/2$ is a fixed point. The recovery applies dephasing again, making the state worse rather than better—confirming that the gravitational dephasing information is genuinely lost and cannot be recovered from the position degree of freedom alone. The recovery fidelity satisfies the channel-setting bound (see Lemma 1):

$$\tau_{\rm Pik} = 1 - F(\rho_0,\, \mathcal{R} \circ \mathcal{N}_p(\rho_0)) \le 1 - e^{-\Sigma_{\rm Pik}}\,, \tag{7}$$

where $\Sigma_{\rm Pik} = H_{\rm bin}\!\left(\frac{1+p}{2}\right) \le \ln 2$ is the binary entropy of the dephasing eigenvalue, and the bound is exactly saturated at $p = 0$ (complete dephasing).

Distinction from $\Sigma_{\rm grav}$.—We stress that the Pikovski entropy production $\Sigma_{\rm Pik}$, bounded above by $\ln 2$, is fundamentally different from $\Sigma_{\rm grav} = -\ln(-g_{00})$, which diverges at a horizon. The Pikovski mechanism produces a *probe-dependent* dephasing channel (depending on internal energy gap ΔE, height separation Δh, and observation time t), while the gravitational channel of Sec. is universal and geometric. The Pikovski mechanism provides physical motivation that gravity produces information-theoretic entropy, but does not itself derive $\Sigma_{\rm grav}$.

Numerical estimate.—The fundamental gravitational scale at height separation Δh on Earth is $g\,\Delta h/c^2 \sim 10^{-19}$ for $\Delta h \sim 1$ mm. The Pikovski decoherence rate is probe-dependent: $\Gamma \sim \Delta E\, g\, \Delta h/(\hbar c^2)$, where ΔE is the internal energy gap. For heavy molecules ($\Delta E/\hbar \sim 10^{12}$ s^{-1}), the decoherence timescale can reach seconds [14]. In the τ framework, the gravitational potential sets a fundamental floor on temporal asymmetry:

$$\frac{g\,\Delta h}{c^2} \sim 10^{-19}\ (\Delta h \sim 1\ {\rm mm})\,, \tag{8}$$

but the actual Pikovski $\tau_{\rm Pik} = (1-p^2)/2$ can be much larger for systems with large $\Delta E\, t/\hbar$. In practice, $\tau_{\rm thermal} \gg \tau_{\rm Pik}$ for all terrestrial systems.

Caveat.—Bonder *et al.* [15] argued that the Pikovski decoherence is an artifact of the coordinate choice and vanishes in a freely falling frame. The debate remains open [16]. In our framework, this corresponds to the question of whether $\mathcal{N}_{\rm grav}$ is a physical channel or a coordinate-dependent one. We note that $\Sigma_{\rm grav}$ as defined in Eq. (5) is manifestly coordinate-dependent (it

TABLE II. Black hole results as instances of the τ bound. Classification: KNOWN = established in the literature; NEW SYNTHESIS = known results reinterpreted through τ; SPECULATIVE = conjectured connections.

Result	τ form	Status
Petz bound	$F \geq e^{-\Sigma/2}$	KNOWN
Complementarity	$\tau_{\rm in}=0$, $\tau_{\rm ext}\sim 1$	NEW SYNTH
Page curve	$\tau(t) \leq 1-e^{-S_P(t)/2}$	NEW SYNTH
HP protocol	HP decoder = Petz map	KNOWN
EW reconstr.	$F^2 \geq e^{-\Delta S_{\rm gen}}$	KNOWN
ER=EPR	ER bridge = zero-τ	SPECUL
Scrambling	$\lvert d\tau/dt\rvert \leq \frac{2\pi}{\beta}\tau$	SPECUL

involves the Newtonian potential), and the covariant formulation via the Killing vector norm $F_{\rm bound}(A \to B) = |\xi|_A/|\xi|_B$ [37] resolves this for stationary spacetimes.

Connection to the Unruh effect.—By the equivalence principle, a uniformly accelerated observer experiences a thermal bath at temperature $T_U = \hbar a/(2\pi c k_{\rm B})$ [20]. This thermal environment generates entropy production $\Sigma > 0$ and hence $\tau > 0$: the Unruh effect creates a gravitational arrow of time for the accelerated observer, even in flat spacetime [KNOWN; NEW in τ language].

BLACK HOLE PHYSICS THROUGH τ

Several established results in black hole physics can be reinterpreted as instances of the τ bound (1). Table II classifies these connections.

Complementarity as observer-dependent τ

Black hole complementarity [31, 32] states that infalling and external observers give *complementary* but non-contradictory descriptions. In our language [NEW SYNTHESIS]:

Infalling observer: The local frame is inertial (equivalence principle), so $\Sigma_{\rm grav,local} = 0$ and $\tau_{\rm in} = 0$—no information loss, no arrow of time, smooth horizon crossing.

External observer: The signal is exponentially redshifted, $\Sigma_{\rm grav} \to \infty$ as $r \to r_{\rm s}$, so $\tau_{\rm ext} \to 1$—near-complete information loss, a strong gravitational arrow of time.

Complementarity is the statement that τ depends on the *partition* (which degrees of freedom are traced out), not on the underlying unitary dynamics. Different observers make different partitions and therefore assign different values of τ to the same process. This is precisely the observer-dependence built into the τ framework: τ is defined relative to a channel $\mathcal{N}$, and different observers define different channels.

The Page curve as $\tau(t)$

The Page curve [23] describes the entanglement entropy $S_{\rm rad}(t)$ of Hawking radiation as a function of time: it rises until the Page time t_P, then decreases to zero as the black hole evaporates.

In the τ framework [NEW SYNTHESIS], the recovery fidelity for the initial state from the radiation satisfies

$$\tau(t) \;\leq\; 1 - e^{-S_P(t)/2}\,, \tag{9}$$

where $S_P(t)$ is the Page entropy $S_P = \min(S_{\rm rad}, S_{\rm BH})$. Before the Page time, S_P is large and $\tau \sim 1$: the initial state is nearly unrecoverable from the radiation. After the Page time, S_P decreases and $\tau \to 0$: recovery becomes progressively easier. Complete evaporation ($S_{\rm BH} = 0$) gives $\tau = 0$: unitarity is restored.

The Page curve is thus a trajectory in τ space, with the Page time marking the transition from $\tau \approx 1$ (information hidden) to $\tau \to 0$ (information recovered). We note that identifying S_P (entanglement entropy) with Σ (QRE decrease) is heuristic: these are distinct information-theoretic quantities that agree qualitatively but differ in general. A rigorous proof would require constructing the Hawking radiation as a quantum channel and computing its entropy production directly.

Hayden–Preskill as τ bound

Hayden and Preskill [24] showed that a diary thrown into an old (post-Page-time) black hole can be recovered from just $O(\log S)$ additional Hawking quanta. Cotler *et al.* [26] proved that the optimal decoder for the Hayden–Preskill protocol *is* the Petz map [KNOWN].

In τ language: after the Page time, $\Sigma_{\rm diary} \sim \log S$ (scrambling time), so $\tau_{\rm diary} \leq 1-e^{-\log S/2} = 1-1/\sqrt{S} \approx 1$ for large S. The diary is formally hard to recover ($\tau \approx 1$), but the *additional* entropy production for the next $O(\log S)$ quanta is small, allowing efficient recovery. The HP decoder being the Petz map means that Petz recovery is not merely a bound but the *actual* recovery strategy.

Entanglement wedge reconstruction

In AdS/CFT, bulk operators within the entanglement wedge of a boundary region R can be reconstructed on R using the Petz map [25, 26]. The reconstruction fidelity satisfies [KNOWN]

$$F^2 \;\geq\; e^{-\Delta S_{\rm gen}}\,, \tag{10}$$

where $\Delta S_{\rm gen}$ is the change in generalized entropy across the RT surface [27, 28]. This is an instance of the Petz bound (1) with $\Sigma = \Delta S_{\rm gen}$.

The τ perspective adds a physical interpretation: $\tau_{\rm holo} = 1 - \sqrt{e^{-\Delta S_{\rm gen}}}$ quantifies how much of the bulk information is *lost* when restricting to a boundary subregion. The RT surface area controls this information loss, giving τ a direct geometric meaning.

ER=EPR as zero-τ channel

Maldacena and Susskind [29] conjectured that Einstein–Rosen bridges and Einstein–Podolsky–Rosen entanglement are equivalent: ER=EPR. In τ language [SPECULATIVE]: an ER bridge connecting two black holes is a channel with $\tau = 0$—information traverses the bridge without loss, consistent with the maximal entanglement between the two sides. The non-traversability of the ER bridge for classical observers reflects the *complexity* of the Petz recovery, not its fidelity: $F = 1$ but the recovery operation requires exponential circuit depth [30].

This suggests a refinement: alongside τ (fidelity loss), one should track the *complexity* of the Petz map. A complexity-weighted τ might distinguish between "easy" recovery ($F \approx 1$, low complexity) and "hard" recovery ($F \approx 1$, high complexity—the ER=EPR regime). We leave this as an open direction.

WHAT τ CAN AND CANNOT EXPLAIN

Can do (with mathematical support):

1. Light speed = recovery fidelity (weak-field exact) — Eqs. (3), (4).
2. Event horizon = $\tau = 1$ (operational definition) — $\Sigma_{\rm grav} \to \infty \Rightarrow F \to 0$.
3. Page curve described by $\tau(t)$ — Eq. (9).
4. HP protocol = τ bound instance — Cotler *et al.* [26].
5. Entanglement wedge reconstruction = τ bound + RT area — Eq. (10).
6. Pikovski decoherence = Petz recovery problem — Sec. .
7. Complementarity = observer-dependent τ.
8. Scrambling rate bound: $|d\tau/dt| \leq (2\pi/\beta)\tau$ [conjectured, from MSS bound [30]].

Cannot do (confirmed limitations of the present paper):

1. Dark matter at the level of Sec. ($\tau_{\rm grav} \sim 10^{-19}$ on Earth: far too small to generate galactic dynamics). At galactic scales, quantum corrections to the gravitational channel produce running G, extending $\Sigma_{\rm grav}$ to include logarithmic corrections (Paper III [38]).
2. Dark energy (does not predict the value of Λ).
3. Derive g_{ab} from τ (the relation $g_{00} \to \Sigma_{\rm grav} \to \tau$ is one-directional; inverting requires additional input).
4. Solve the horizon problem (requires causal connection, which τ does not provide).
5. $c_{\rm eff} = c(1-\tau)$ as an exact identity (it is a first-order approximation valid for $\Sigma_{\rm grav} \ll 1$).
6. Macroscopic arrow of time from gravitational decoherence ($\tau_{\rm grav} \ll \tau_{\rm thermal}$).

DISCUSSION AND OUTLOOK

Open question 1: τ at the Planck scale.—At the Planck length, $\Sigma_{\rm grav} \sim l_P/l_P = O(1)$, so $\tau \sim O(1)$. This suggests that Planck-scale physics lies at the boundary between "recoverable" and "irrecoverable" information—a natural setting for quantum gravity. Whether $\tau = 1$ (complete loss) is achievable at finite $\Sigma_{\rm grav}$ is a question about the tightness of the Petz bound.

Open question 2: Complexity-weighted τ.—The ER=EPR discussion (Sec.) suggests that τ alone is insufficient: one needs the complexity of the Petz map as an additional variable. This may connect to the firewall debate [33]: if the Petz recovery exists ($F > 0$) but requires trans-Planckian complexity, is information "really" recoverable?

Open question 3: Non-Markovian gravity.—In quantum channels, $\Sigma < 0$ corresponds to information backflow (non-Markovian dynamics), giving $\tau_{\rm signed} < 0$ (time reversal) [2]. In gravitational physics, $\Sigma_{\rm grav} < 0$ would require $|\Phi| < 0$—repulsive gravity. Whether cosmological scenarios (e.g., dark energy-dominated expansion) can produce gravitational backflow is an open question.

Pikovski Petz recovery: a completed calculation.—The explicit Petz map for the Pikovski dephasing channel $\mathcal{N}_p$ has been constructed [37]: with reference state $\sigma = I/2$, the Petz map equals $\mathcal{N}_p$ itself (Sec.), giving $F_{\rm Petz} = (1 + p^2)/2$. The channel-setting bound $F \geq e^{-\Sigma_{\rm Pik}}$ is satisfied for all p and exactly saturated at $p = 0$ (complete dephasing). The stronger CMI bound $F \geq e^{-\Sigma/2}$ is not achieved, consistent with the dephasing channel being self-adjoint—the rotated Petz map reduces to the standard Petz map, and no rotation can improve recovery.

Broader perspective.—The identification $c_{\text{eff}}/c = F_{\text{bound}}$ connects three domains that are usually treated separately: quantum information (Petz recovery), thermodynamics (entropy production), and gravitational physics (the metric). This resonates with the Jacobson program [22], which derives Einstein's equations from thermodynamic considerations; our approach is complementary, identifying the metric component g_{00} directly with recovery fidelity. The τ framework provides a common language: $\tau = 0$ is flat spacetime, reversible dynamics, and perfect retrodiction; $\tau > 0$ is curved spacetime, irreversible dynamics, and imperfect retrodiction; $\tau = 1$ is an event horizon, complete decoherence, and total information loss.

Connection to the weak-field program.—At galactic scales, quantum corrections to the gravitational channel produce a running gravitational coupling, extending Σ_{grav} to include logarithmic corrections that modify galactic dynamics (Paper III [38]). The Khronon conservation law $I_0 = \text{const}$ provides the cosmological counterpart of Σ_{grav} [38], unifying the strong-field regime explored here with the weak-field phenomenology of dark matter. In the Blanchet–Skordis Khronon framework [38], the inverse lapse $Q = 1/N_\phi$ of the Khronon foliation provides a covariant generalization: $\Sigma_{\text{grav}} = 2 \ln Q$ on any background in the adiabatic regime, with $Q = 1/\sqrt{-g_{00}}$ recovering the static result of Sec. .

Whether this correspondence extends beyond stationary spacetimes—to cosmology, to quantum gravity, to the measurement problem—remains to be seen.

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[1] The bound $F \geq e^{-\Sigma/2}$ holds in the conditional mutual information (CMI) setting of Fawzi–Renner [4], where $\Sigma = I(A{:}C|B)$ involves a tripartite state. In the channel divergence setting—$\Sigma = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$—the universal bound is $F \geq e^{-\Sigma}$ [6]. The factor of 2 arises because the CMI is a double relative entropy. In this paper we define $\Sigma_{\text{grav}} \equiv 2|\Phi|/c^2$ so that the identification $c_{\text{eff}}/c = e^{-\Sigma_{\text{grav}}/2}$ takes the CMI form; equivalently, in the channel setting one writes $c_{\text{eff}}/c = e^{-\Sigma_{\text{ch}}}$ with $\Sigma_{\text{ch}} = |\Phi|/c^2 = \Sigma_{\text{grav}}/2$.

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